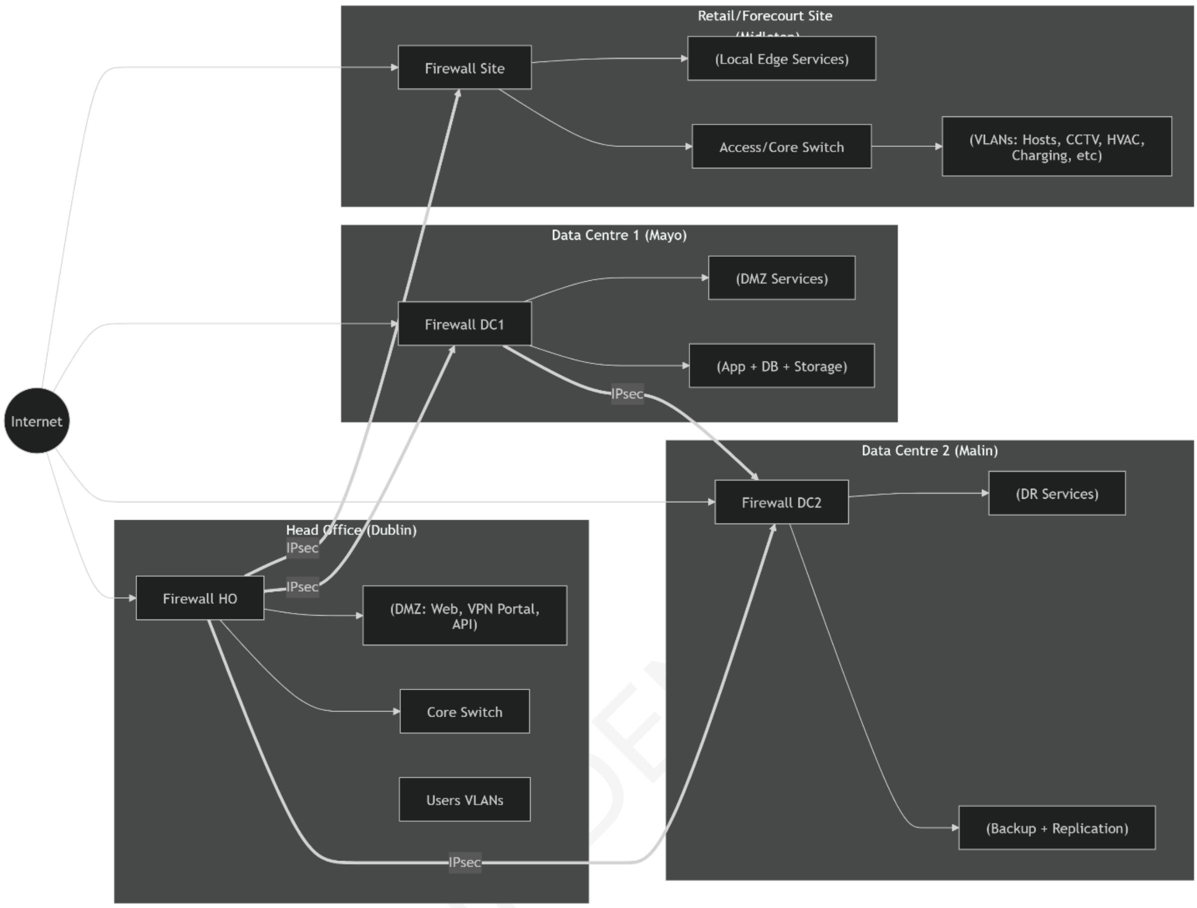
Network Architecture Diagram (Markdown)



GNS3 TOPOLOGY (DEVICE-BY-DEVICE LAYOUT)

Core Devices Per Site

Head Office (HO)

- 1 × Cisco ASA (HO-FW)
- 1 × L3 Switch (HO-Core)
- 1 × L2 Access Switch
- 1 × Web Server (DMZ)
- 1 × Test PC

Data Centre 1 (DC1)

- 1 × Cisco ASA (DC1-FW)

- 1 × L3 Switch
- 2 × Servers (App + DB)

Data Centre 2 (DC2)

- 1 × Cisco ASA (DC2-FW)
- 1 × L3 Switch
- 2 × Servers (Backup + DR)

Retail Site (Midleton)

- 1 × Cisco ASA (SITE-FW)
- 1 × L3 Switch (Site-Core)
- 2 × L2 Access Switches
- End Devices:
    - PC (Hosts VLAN)
    - CCTV device
    - EV charger simulator (loopback / Linux VM)
    - HVAC device

Internet Simulation

- 1 × Cloud node OR
- 1 × Router (ISP)

INTERFACE PLAN

Interface	Purpose	IP
g0/0	outside	DHCP / Public
g0/1	inside	10.x.x.1
g0/2	dmz	172.16.x.1

CONFIGS (PER NODE)

HEAD OFFICE FIREWALL (ASA)

hostname HO-FW

interface g0/0

nameif outside

security-level 0

ip address 203.0.113.1 255.255.255.0

interface g0/1

nameif inside

security-level 100

ip address 10.0.0.1 255.255.0.0

interface g0/2

nameif dmz

security-level 50

ip address 172.16.0.1 255.255.255.0

! NAT

object network INSIDE

subnet 10.0.0.0 255.255.0.0

nat (inside,outside) dynamic interface

! DMZ Web Server

object network WEB

host 172.16.0.10

nat (dmz,outside) static 203.0.113.10

access-list OUTSIDE-IN permit tcp any host 203.0.113.10 eq 443

access-group OUTSIDE-IN in interface outside

VPN (HO side)

```
crypto isakmp policy 10
encryption aes
hash sha256
authentication pre-share
group 14

crypto ipsec transform-set ESP-AES-SHA esp-aes esp-sha-hmac

access-list VPN_SITE permit ip 10.0.0.0 255.255.0.0 10.50.0.0 255.255.0.0

crypto map VPN 10 match address VPN_SITE
crypto map VPN 10 set peer 203.0.113.50
crypto map VPN 10 set transform-set ESP-AES-SHA

crypto map VPN interface outside

tunnel-group 203.0.113.50 type ipsec-l2l
tunnel-group 203.0.113.50 ipsec-attributes
pre-shared-key cisco123
```


SITE FIREWALL (ASA)

hostname SITE-FW

```
interface g0/0
nameif outside
security-level 0
ip address 203.0.113.50 255.255.255.0

interface g0/1
```

```
nameif inside
security-level 100
ip address 10.50.0.1 255.255.0.0

interface g0/2
nameif dmz
security-level 50
ip address 172.16.50.1 255.255.255.0

! NAT
object network SITE-NET
subnet 10.50.0.0 255.255.0.0
nat (inside,outside) dynamic interface

! VPN
access-list VPN-HO permit ip 10.50.0.0 255.255.0.0 10.0.0.0 255.255.0.0

crypto map VPN 10 match address VPN-HO
crypto map VPN 10 set peer 203.0.113.1
crypto map VPN 10 set transform-set ESP-AES-SHA
crypto map VPN interface outside

tunnel-group 203.0.113.1 type ipsec-l2l
tunnel-group 203.0.113.1 ipsec-attributes
pre-shared-key cisco123
```


DATA CENTRE FIREWALL (DC1 Example)

```
hostname DC1-FW
```

interface g0/0

nameif outside

ip address 203.0.113.11 255.255.255.0

security-level 0

interface g0/1

nameif inside

ip address 10.1.0.1 255.255.0.0

security-level 100

! NAT

object network DC1-NET

subnet 10.1.0.0 255.255.0.0

nat (inside,outside) dynamic interface

L3 SWITCH (SITE CORE)

hostname SITE-CORE

ip routing

vlan 2

name HOSTS

vlan 4

name POWER

vlan 8

name HVAC

vlan 11

name CCTV

interface vlan 2

ip address 10.50.2.1 255.255.255.0

```
interface vlan 4
ip address 10.50.4.1 255.255.255.0

interface vlan 8
ip address 10.50.8.1 255.255.255.0

interface vlan 11
ip address 10.50.11.1 255.255.255.0

! Default route to firewall
ip route 0.0.0.0 0.0.0.0 10.50.0.1
```


ACCESS SWITCH

```
hostname ACCESS-SW1

interface range fa0/1-10
switchport mode access
switchport access vlan 2

interface range fa0/11-15
switchport mode access
switchport access vlan 11

interface fa0/24
switchport trunk encapsulation dot1q
switchport mode trunk
```

TEST PLAN

Internet

ping 8.8.8.8

VPN

ping 10.0.0.10 (from site)

show crypto ipsec sa

` `

DMZ

curl https://203.0.113.10

Technical Design Notes (Markdown)

Location	CIDR
Head Office	10.0.0.0/16
Data Centre 1	10.1.0.0/16
Data Centre 2	10.2.0.0/16
Site (Midleton)	10.50.0.0/16

Perimeter Security Zones

one	Purpose
Outside	Internet
Inside	Internal VLANs
DMZ	Public-facing services
VPN	Encrypted tunnels

DMZ Services

Location	Services
Head Office	VPN portal, Web apps, API gateway
DC1	Primary app + database frontends
DC2	DR web endpoints
Site	EV charging gateway, CCTV relay

Core VLAN Mapping (Site Example)

VLAN	Function
10.0.2.0	Hosts
10.0.4.0	Power infrastructure (chargers)
10.0.6.0	Servers
10.0.9.0	Storage
10.0.10.0	NetMan
10.0.11.0	CCTV
10.0.8.0	HVAC

Routing Design

Dynamic routing (OSPF or EIGRP) inside sites

Static routes or BGP between firewalls

Default route → Internet via firewall

CLI CONFIG (DIFF STYLE)

Site-to-Site VPN (IPsec)

Head Office Firewall (HO_FW)

```
+ crypto isakmp policy 10
+  encryption aes
```

+ hash sha256

+ authentication pre-share

+ group 14

+ lifetime 86400

+ crypto ipsec transform-set ESP-AES-SHA esp-aes esp-sha-hmac

+ access-list VPN_DC1 extended permit ip 10.0.0.0 255.255.0.0 10.1.0.0 255.255.0.0

+ access-list VPN_DC2 extended permit ip 10.0.0.0 255.255.0.0 10.2.0.0 255.255.0.0

+ access-list VPN_SITE extended permit ip 10.0.0.0 255.255.0.0 10.50.0.0 255.255.0.0

+ crypto map VPN 10 match address VPN_DC1

+ crypto map VPN 10 set peer <DC1_PUBLIC_IP>

+ crypto map VPN 10 set transform-set ESP-AES-SHA

+ crypto map VPN 20 match address VPN_DC2

+ crypto map VPN 20 set peer <DC2_PUBLIC_IP>

+ crypto map VPN 20 set transform-set ESP-AES-SHA

+ crypto map VPN 30 match address VPN_SITE

+ crypto map VPN 30 set peer <SITE_PUBLIC_IP>

+ crypto map VPN 30 set transform-set ESP-AES-SHA

+ crypto map VPN interface outside

NAT + Internet Access

+ object network INSIDE-NET

+ subnet 10.0.0.0 255.0.0.0

+ nat (inside,outside) dynamic interface

+ route outside 0.0.0.0 0.0.0.0 <ISP_GATEWAY>

DMZ Configuration

+ interface GigabitEthernet0/2

+ nameif dmz

+ security-level 50

+ ip address 172.16.1.1 255.255.255.0

+ object network WEB-SERVER

+ host 172.16.1.10

+ nat (dmz,outside) static <PUBLIC_IP>

+ access-list OUTSIDE-IN permit tcp any host <PUBLIC_IP> eq 443

+ access-group OUTSIDE-IN in interface outside

Inter-VLAN Control (Site Firewall)

+ access-list INSIDE-CONTROL deny ip 10.50.4.0 255.255.255.0 10.50.2.0 255.255.255.0

+ access-list INSIDE-CONTROL permit ip any any

+ access-group INSIDE-CONTROL in interface inside

` `

Basic Testing Commands

show crypto isakmp sa

show crypto ipsec sa

ping 10.1.0.10

traceroute 10.2.0.10

show nat

` `

Mobile & Contractor Secure Access

Requirements

External users need access to:

- Applications
- Maintenance systems
- Possibly EV infrastructure

Secure Access Design

Remote Access VPN (SSL VPN)

- Terminate on **Head Office firewall**
- Technologies:
  - Cisco AnyConnect / OpenVPN
- Features:
  - MFA (Azure AD / RADIUS)
  - Device posture check

Zero Trust Principles

Control	Implementation
Identity-based access	Azure AD / IAM
Least privilege	VLAN / firewall ACLs
Device posture	Endpoint compliance
Session monitoring	Logging & SIEM

Role-Based Access

User Type	Access
Staff	Full internal apps

Contractors	Restricted VLAN or jump host
Maintenance	ONLY EV charger / HVAC systems

Jump Host (Critical)

- Located in DMZ or secure VLAN
- Contractors:
  - VPN → Jump Host → Target systems
- NO direct access to core network

Application Proxy (Optional but Strong)

- Expose apps via:
  - Reverse proxy (NGINX)
  - Azure App Proxy / Zscaler

Network Segmentation

From your VLAN table:

- EV charging → isolated
- CCTV → isolated
- HVAC → isolated

Firewall allows ONLY required flows

Example Access Flow

Contractor Laptop - SSL VPN (MFA) - DMZ Jump Server - Restricted VLAN (e.g., HVAC / Chargers only)

Summary

This design delivers:

✓ Site-to-site VPN (HO ↔ DC1 ↔ DC2 ↔ Site)
✓ Internet access via NAT
✓ Segmented DMZ services
✓ Strong VLAN isolation (based on your spreadsheet)
✓ Secure contractor/mobile access via VPN + Zero Trust